



Course: Database Administration

Lecturer: Mr. Thear Sophal

Week 8 – Practical Session

Database Server - Hosting

Part 1: Verify Aiven Database Hosting

Tasks

1. Create or log in to your Aiven account.
2. Create a MySQL database instance.
3. Import the provided `hr_db` database.
4. Connect to the database using DBeaver.
5. Execute at least 5 SQL queries to verify the connection.
6. Capture screenshots showing:
 - Aiven service dashboard
 - Database connection settings
 - DBeaver connected successfully
 - Query results

Part 2: Host Your Local Database Using ngrok

Tasks

1. Install ngrok on your computer.
2. Ensure your MySQL server is running locally.
3. Expose your MySQL port using:

ngrok tcp 3306

4. Record:
 - Public host address
 - Public port number
5. Connect to the ngrok endpoint using DBeaver.
6. Execute test queries on the exposed database.
7. Test at least one user account created in previous lab sessions.

Part 3: Explore Alternative Hosting Providers

Tasks

Research at least **three** hosting platforms from the list below:

- Aiven
- Railway
- Render
- PlanetScale
- ScaleGrid
- AWS RDS
- Google Cloud SQL
- Azure Database for MySQL

Create a comparison table containing:

Platform	Free Tier	Supported Database	Ease of Use	Recommended for Student Project
Example: Aiven	Yes	MySQL	Easy	Yes

Questions

1. Which platform is easiest to use?
2. Which platform offers the best free plan?
3. Which platform would you choose for your Final Project?
4. Explain your choice.

Part 4: Final Project Hosting Decision

Tasks

Each project team must decide:

1. Which database hosting platform will be used.
2. Why was the platform selected?
3. Estimated monthly cost (if any).
4. Advantages and limitations.

Deliverable

Prepare a **1-page report** containing:

- Project title
- Team members
- Selected hosting platform
- Justification
- Deployment plan

Submission Requirements

Submit a PDF report containing:

1. Aiven deployment evidence.
2. ngrok deployment evidence.
3. Hosting platform comparison table.
4. Final project hosting decision (for now).

File Naming Format

DBA_Lab7_GroupName.pdf

Submission Deadline

Before the beginning of next week's lab session.

Bonus Challenge (+5 Marks)

Deploy a simple web application (Flask, PHP, Node.js, etc.) that connects to your hosted database and demonstrates:

- Database connection
- Data retrieval
- Basic CRUD operations